

SHAKETABLE PROJECT

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INTRODUCTION

- I'm from Tokyo, Japan
- 18.5% of M6 or higher earthquakes happen in Japan.
(Japan takes up 0.25–0.29 % of world area)
- The Great East Japan Earthquake (2011)

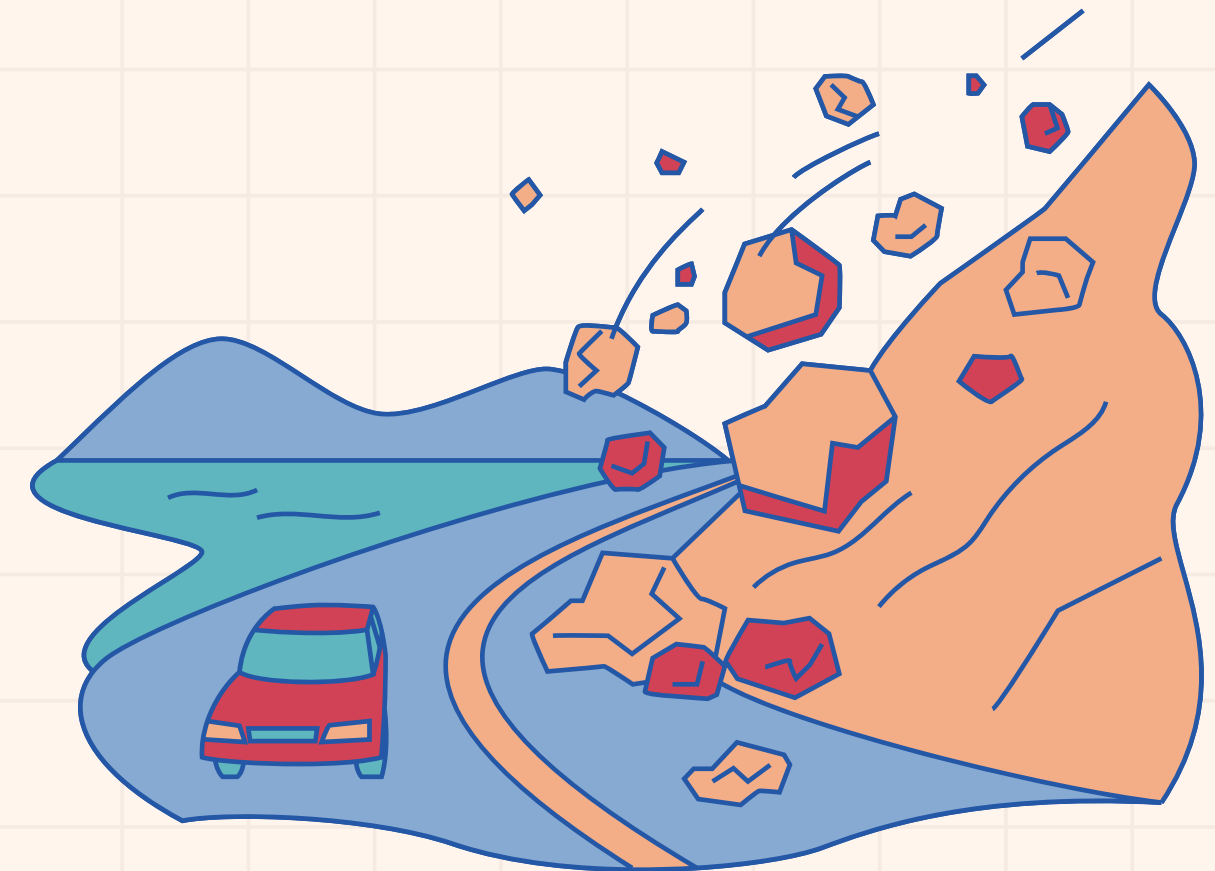
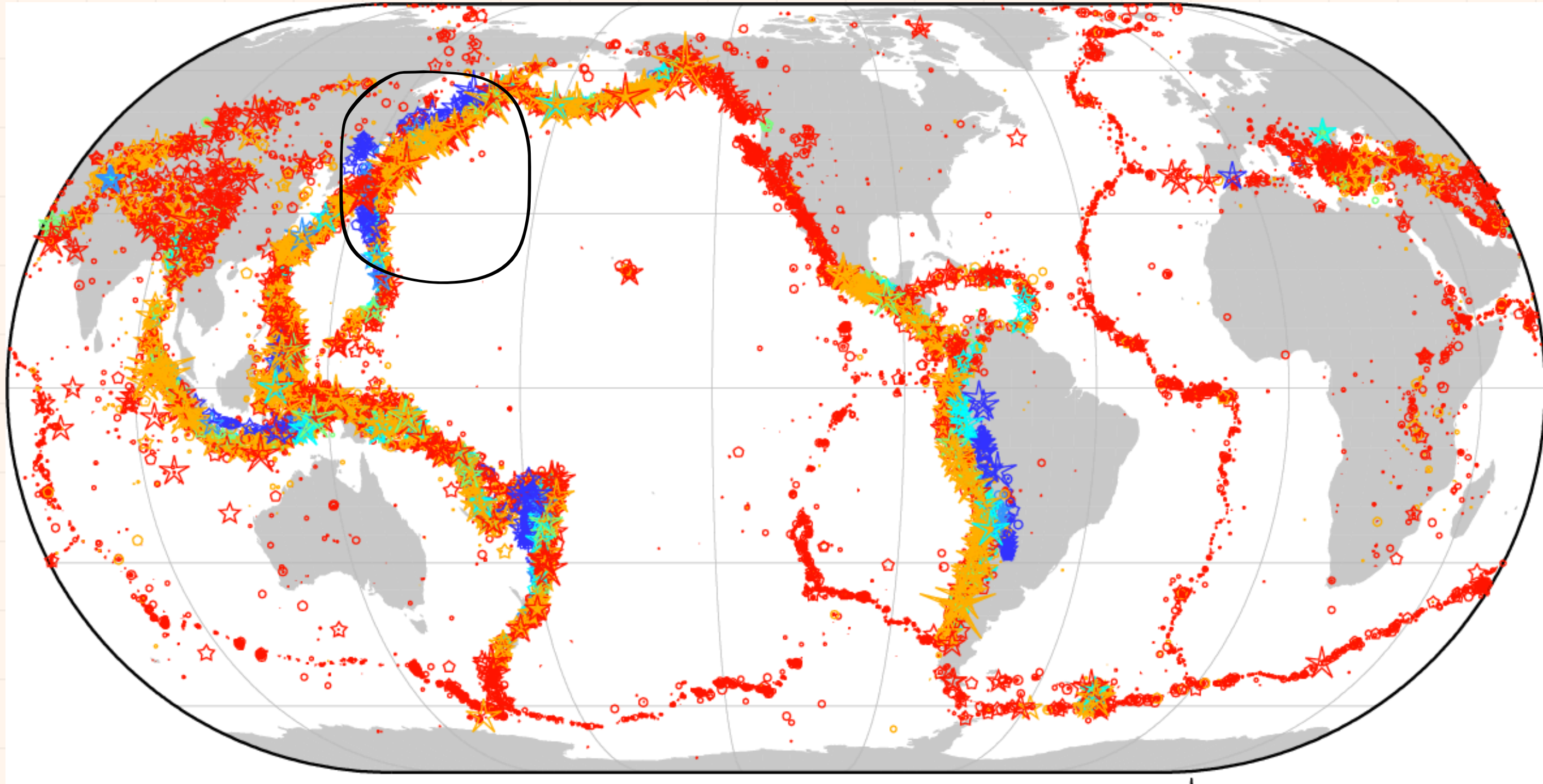


PLATE BOUNDARIES



https://www.maxapress.com/app/figure_table/66433dc0fa6c58429b9216e3

DEATH CAUSE

Tsunami



https://www.bousai.go.jp/kohou/kouhoubousai/h23/63/special_01.html

Nuclear pollution



<https://www.jacom.or.jp/nousei/closeup/2022/220311-57469.php>

Collapsing



<https://kahoku.news/articles/20220314khn000019.html>

Cracks



https://www.jiji.com/jc/d4?p=heq117-915826&d=d4_oo

EXISTING SOLUTION TO EARTHQUAKES



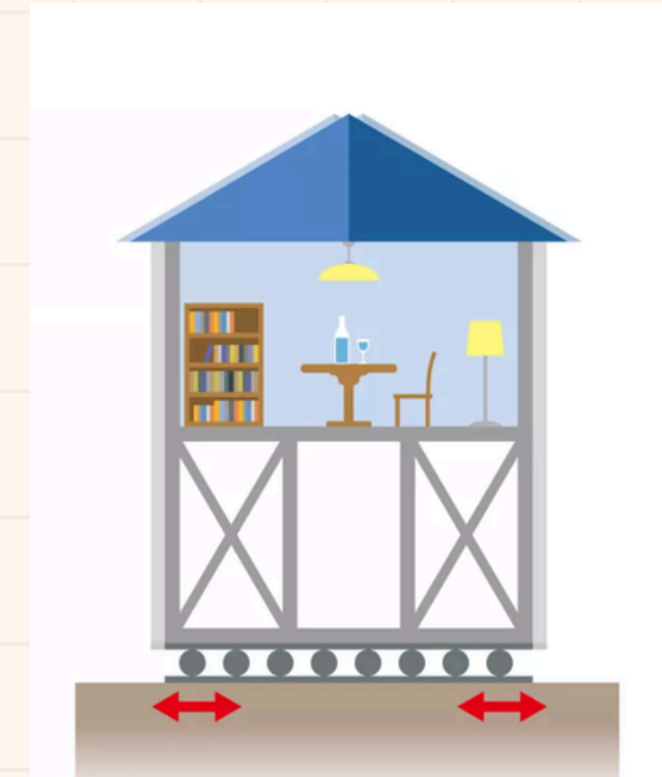
Earthquake resistant:

- stronger walls & structures
- low cost, low safety



Damping:

- absorbing movement with dampers
- medium cost, medium safety



Seismic isolation:

- separating the ground and the architecture
- high cost, high safety

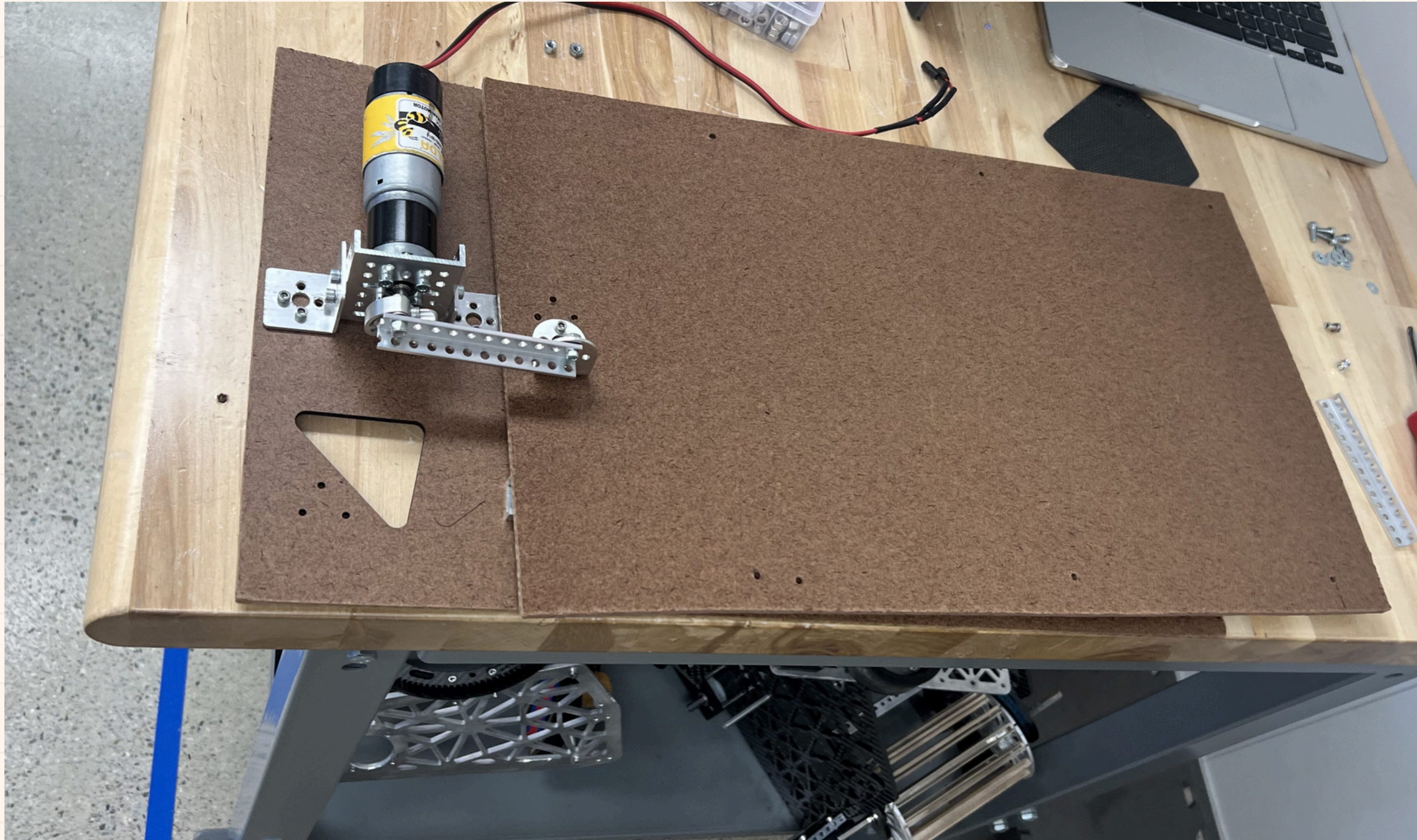
I WANT TO TEST DIFFERENT STRUCTURES!

MATERIALS & METHOD

- boards
- motor
- slide rails
- screws
- metal brackets



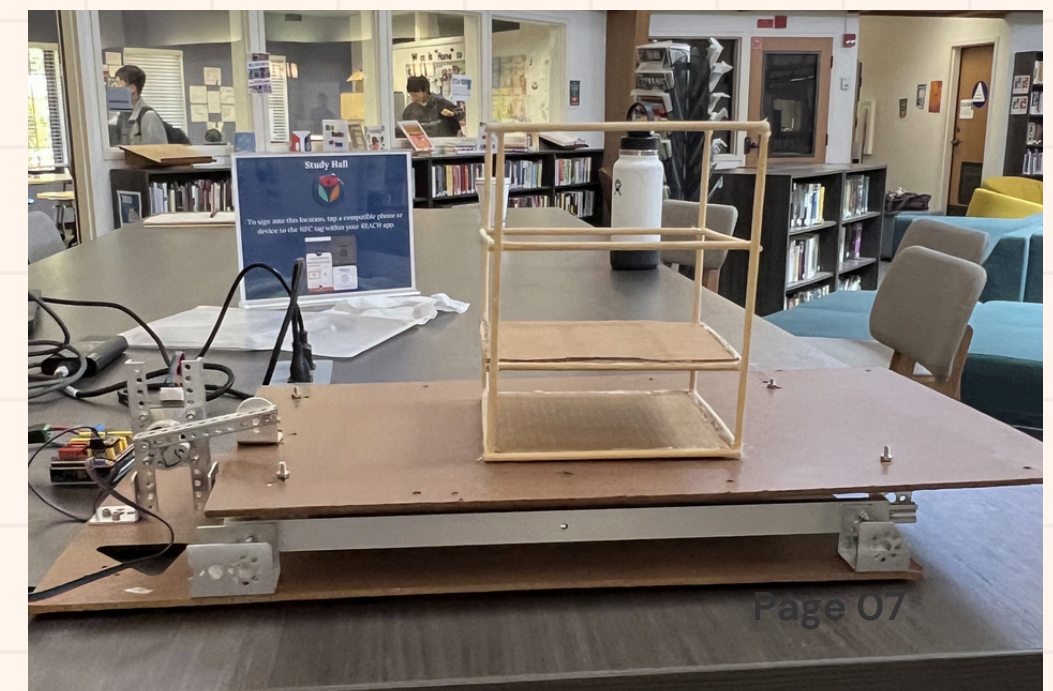
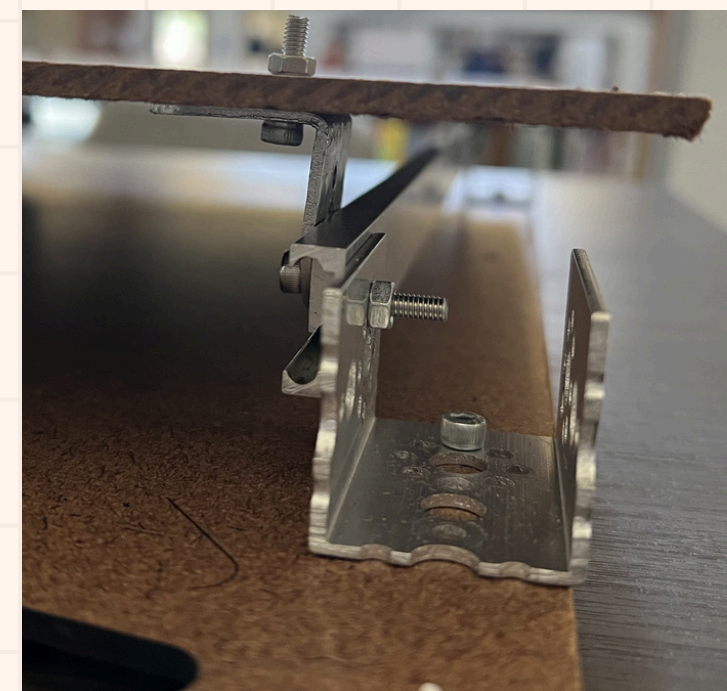
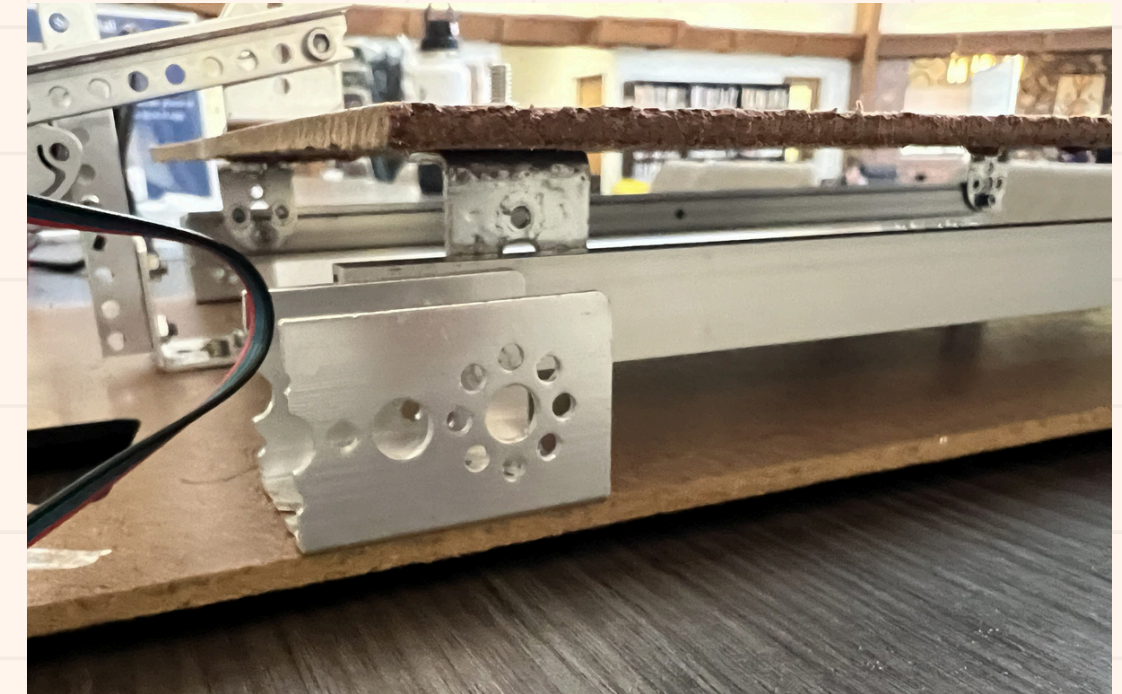
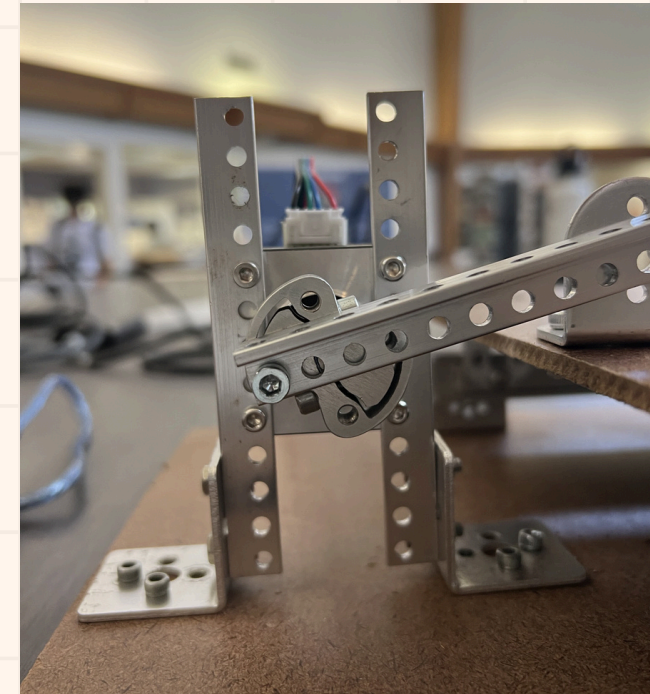
FIRST PROTOTYPE



PROBLEM & SOLUTION

- Friction
 - use the slide rail vertically to create more space in between boards
- Can't control the shake frequency
 - Use stepper motor
- The screws unscrew because of rotation
 - program to go back and forth instead of rotating

SECOND PROTOTYPE



SECOND PROTOTYPE

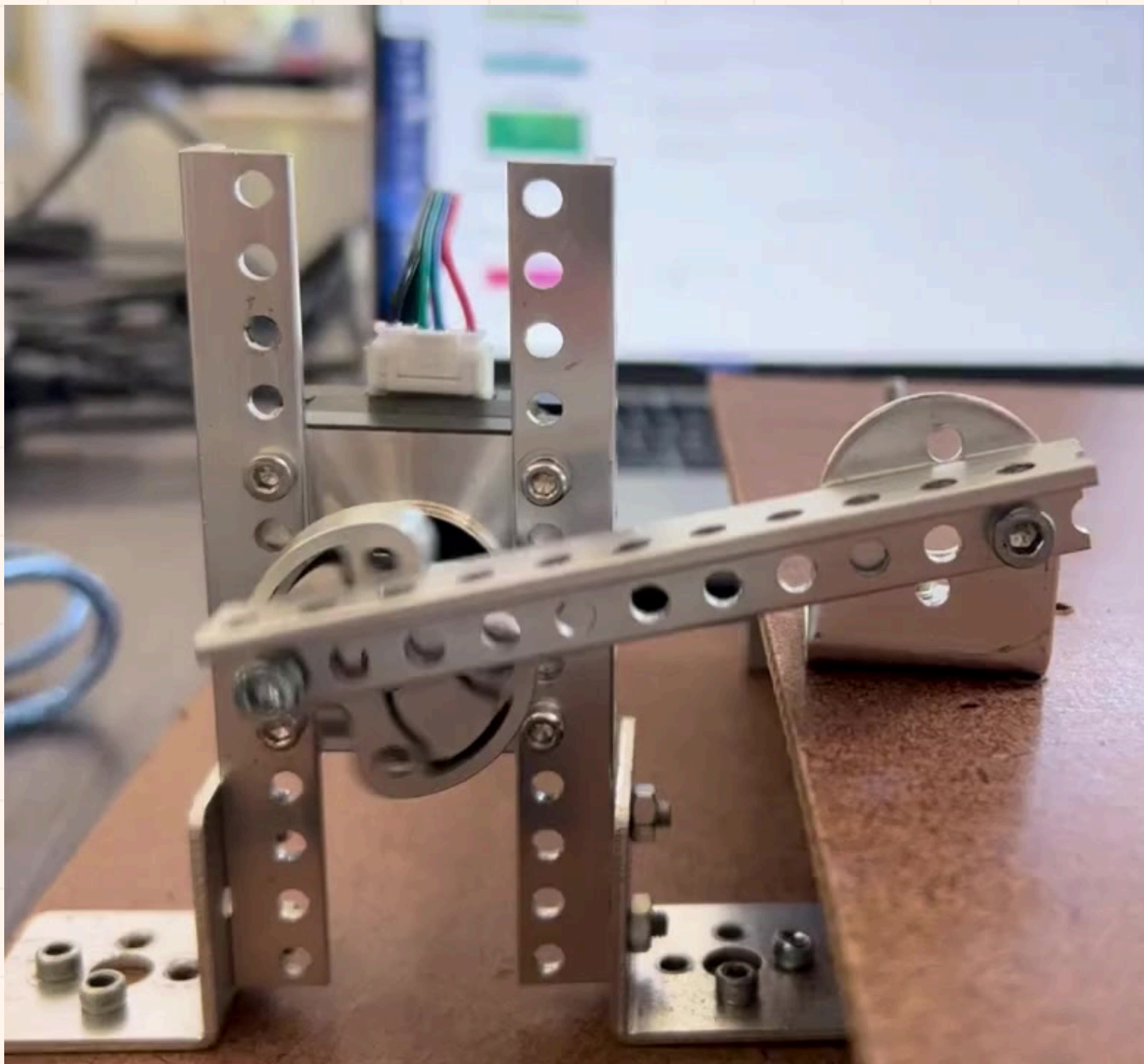
stepper_sketch_earthquake_new.ino

```
1  int ENABLE_PIN=8;
2  int M1_STEP_PIN = 3 ;
3  int M1_DIR_PIN = 6 ;
4  // int M2_STEP_PIN = 2
5  // int M1_DIR_PIN = 5
6
7  int delay_length= 100;
8  int steps_per_direction = 100;
9  int delay_between_steps = 100 ;
10
11 void setup() {
12     pinMode (M1_STEP_PIN, OUTPUT);
13     pinMode (M1_DIR_PIN, OUTPUT);
14     pinMode (ENABLE_PIN, OUTPUT);
15 }
16
```

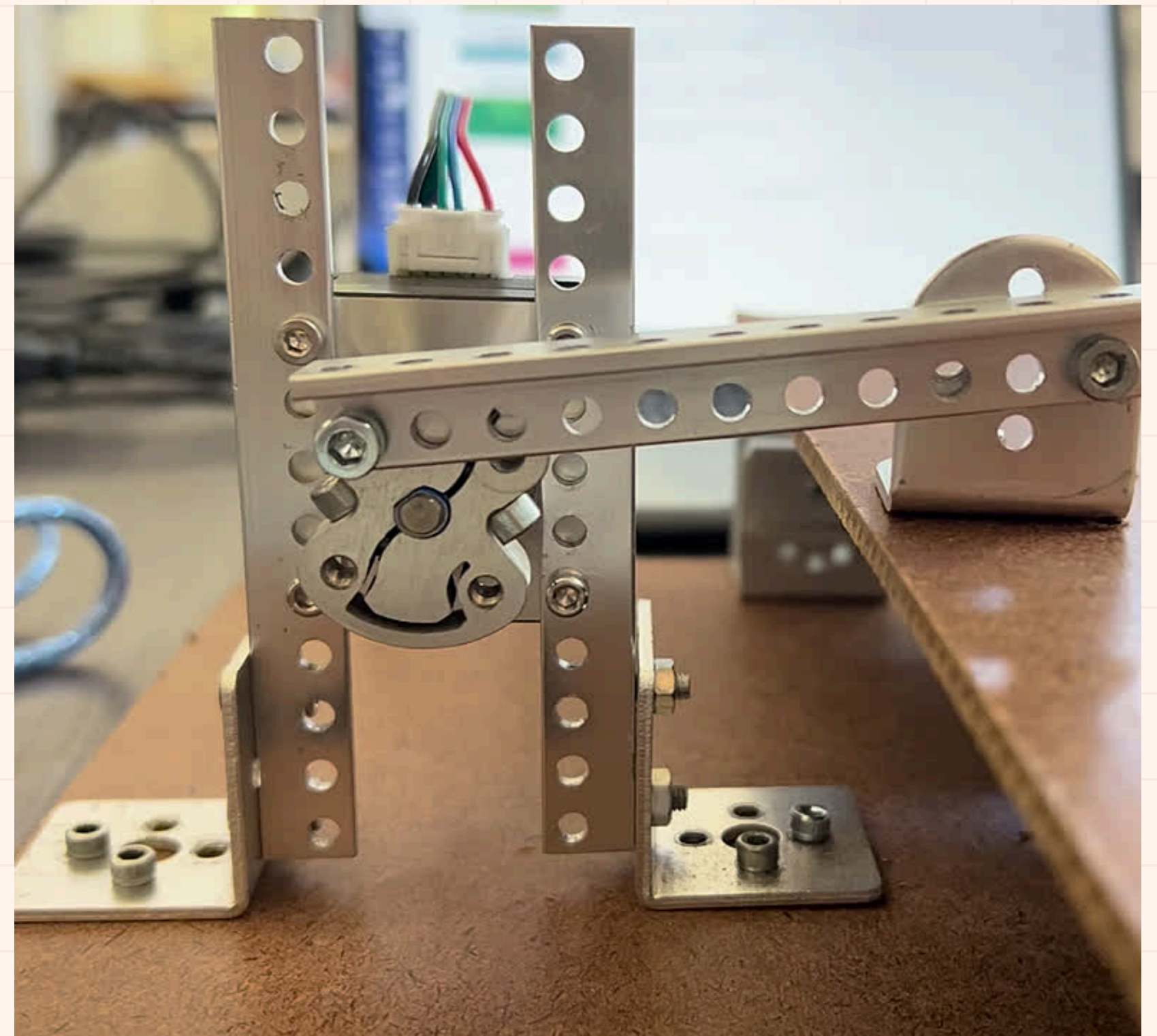
adjusting the delay length to
recreate the shake similar to
earthquake

```
16
17 void loop() {
18     digitalWrite(M1_DIR_PIN, LOW);
19
20     for (int i=0; i<steps_per_direction; i++){
21         digitalWrite(M1_STEP_PIN, HIGH);
22         delayMicroseconds(delay_length);
23         digitalWrite(M1_STEP_PIN, LOW);
24         delayMicroseconds(delay_length);
25     }
26     delay(delay_between_steps);
27
28     digitalWrite(M1_DIR_PIN, HIGH);
29
30     for (int i=0; i<steps_per_direction; i++){
31         digitalWrite(M1_STEP_PIN, HIGH);
32         delayMicroseconds(delay_length);
33         digitalWrite(M1_STEP_PIN, LOW);
34         delayMicroseconds(delay_length);
35     }
36     delay (delay_between_steps);
37 }
38
```





wrong placement → doesn't shake



right placement → does shake

THANK YOU

